

Gas Optimization Report

Summary

This report documents expected gas-sensitive areas and the required before/after benchmark for the inline Yul optimization. The actual numeric values should be regenerated on the final commit with `forge snapshot` and pasted into this report before submission.

Optimization target

The AMM quote calculation is used in swaps and frontend quoting. The project includes two equivalent functions:

- `AMMMath.quoteOutSolidity`
- `AMMMath.quoteOutYul`

The test suite checks that both return the same value for normal inputs and fuzzed inputs. The Yul function removes some high-level Solidity overhead in the arithmetic path.

Benchmark command

```
forge snapshot --match-contract GameFiUnitTest --snap docs/.gas-snapshot
forge test --gas-report
```

Before / after table

Function	Implementation	Gas before	Gas after	Result
AMM quote	Solidity arithmetic	TBD	N/A	Baseline
AMM quote	Inline Yul arithmetic	N/A	TBD	Expected lower gas
AMM swap	Uses quote path	TBD	TBD	Compare after implementation
Craft recipe	Resource burn + item mint	TBD	TBD	No Yul used
Vault deposit	ERC-4626 deposit	TBD	TBD	QZ baseline
Rental claim	Pull-payment ETH claim	TBD	TBD	CEI chosen over micro-optimization

L1 vs L2 gas comparison

A separate table is in `docs/L1_L2_GAS_COMPARISON.md`. It must include at least six operations measured on Ethereum Sepolia and the chosen L2 testnet.

Required rows:

1. Craft recipe #1
2. Open loot box request
3. Fulfill loot drop callback
4. AMM add liquidity
5. AMM swap
6. ERC-4626 vault deposit
7. Governor castVote

Optimization decisions

Keep SafeERC20 despite small overhead

SafeERC20 is required for correctness across ERC-20 implementations. The slight gas overhead is justified because unchecked ERC-20 return values are a known source of token integration bugs.

Use uint112 reserves

AMM reserves are stored as uint112, similar to established constant-product AMM designs. This reduces storage footprint and is sufficient for expected resource token supplies.

Avoid over-optimizing governance

Governor and Timelock are not hot paths. The project favors standard OpenZeppelin components over custom gas-optimized governance code.

Use pull payments for rentals

Pull payments may require an extra transaction, but they reduce risk and simplify state transitions. Security is prioritized over minimizing the number of calls.

Use UUPS only for parameters

Only the parameter module is upgradeable. This avoids proxy overhead on high-frequency AMM, crafting, and vault operations.

Final action items

- [] Run `forge snapshot` on the final commit.
- [] Paste the gas snapshot into this document or link `docs/.gas-snapshot`.
- [] Fill before/after numbers for Yul and Solidity quote paths.
- [] Fill L1 vs L2 table with explorer transaction links.